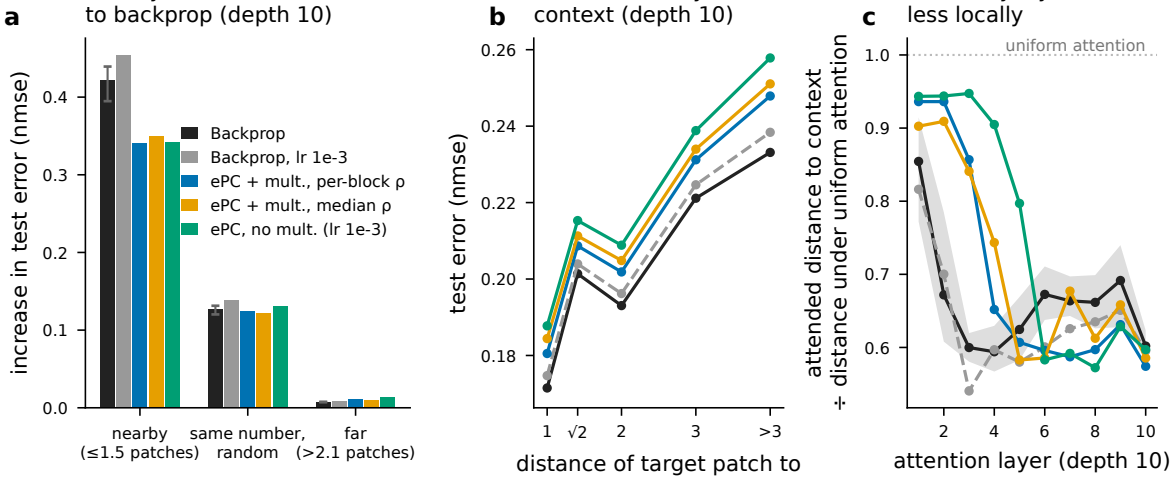
a
Nearby context matters more to backprop (depth 10)
increase in test error (nmse)
0.0
0.1
0.2
0.3
0.4
Backprop
Backprop, lr 1e-3
ePC + mult., per-block ρ
ePC + mult., median ρ
ePC, no mult. (lr 1e-3)
nearby (≤1.5 patches)
same number, random
far (>2.1 patches)
b
Error rises away from context (depth 10)
test error (nmse)
0.18
0.20
0.22
0.24
0.26
1
√2
2
3
>3
distance of target patch to nearest visible context patch
c
PC's early layers attend less locally
uniform attention
attended distance to context ÷ distance under uniform attention
0.6
0.7
0.8
0.9
1.0
2
4
6
8
10
attention layer (depth 10)